



Society of Petroleum Engineers

SPE-227061-MS

New HVFR Fluid Technology for Enhanced Proppant Transport in Varying Salinity Formations: An Unconventional Hydraulic Fracturing Application

D. Wang, Baker Hughes, Beijing, China; T. Ban, CNPC Daqing Oil Field, Beijing, China; Y. Li, Baker Hughes, Beijing, China; N. Spurr, Baker Hughes, Tomball, TX, USA

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227061-MS](https://doi.org/10.2118/227061-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

A traditional fracturing job with slickwater has been a popular choice due to low fluid cost, ability to pump at high rates, and thus allowing for complex fracture geometry, as well as a high regained conductivity of created fractures. However, these fluids need a very high amount of fresh water to deliver a required amount of proppant because the proppant carrying capacity of low viscosity fluid is much lower. The biggest disadvantage in using slickwater is that produced water cannot be used as a fracturing fluid as the polymer is not hydrated properly and fluid stability is low in high salinity water. A high viscosity fluid like a guar or polymer gel can be used but the desired fracture complexity is hard to achieve, and significant formation damage is usually caused leading to low fracture conductivity. A High Viscosity Friction Reducer (HVFR) system is used as a specialized fracturing fluid to reduce friction within the wellbore while allowing for superior proppant transport. The reduced fluid resistance in the formation also helps to create wider, more conductive fractures with more efficient placement of proppant.

In this work, a new HVFR system is developed that can be customized for given water quality, with a wide range of total dissolved solids (TDS) values. This provides a flexibility to design fracturing jobs in unconventional reservoirs based on field conditions. During the development of this fluid, extensive laboratory testing of fluid rheology and performance was done. Friction reduction of the HVFR fluid system was measured for varying TDS values and it was found that friction reduction was >70% for both fresh water and high TDS water (>200,000 ppm) cases. In addition, the hydration rate of the polymer was found to be very fast (<0.5 min). The fluid viscosity profiles as a function of shear rate (3-600 rpm) and HVFR concentration (1-8 gpt (gallon per thousand gallon)) were also established. It was found that viscosity values varied from 700 – 250 cP as the shear rate increased from 3 to 600 rpm for HVFR concentration of 8 gpt. Because of the unique chemistry of the HVFR polymer, even high concentrations (>4 gpt) were found to be cost-effective solutions.

The HVFR system developed was used in several horizontal wells using flowback water. It was found to perform exceedingly well for high TDS water as compared to other fracturing fluids available.

Introduction

Coalbed methane (CBM), also known as coal seam gas, refers to unconventional natural gas that exists as free gas and adsorbed gas in underground coal seams. Coalbed methane includes both shallow coalbed methane (shallow CBM) and deep coalbed methane (deep CBM).

China has abundant coalbed methane resources. With the basic completion of shallow CBM, exploration and development of coalbed methane at depths of 1,000 meters or more has become an inevitable trend. The geological resource volume of coalbed methane at depths ranging from 1,000 to 1,500 meters is 10,611.73 billion cubic meters, and the recoverable resource volume is 4,599.16 billion cubic meters, accounting for 42.31% of the national total. The first deep coalbed methane well in China, Ordos Basin's Ji-Deep 6-7-Ping-01 well, has accumulated gas production exceeding 20,000,000 cubic meters. The deep D6-7P1 horizontal well has a test production rate of over 90,000 cubic meters per day. Ji-Deep 14-5-Ping02 well has accumulated gas production of 26,019,000 cubic meters since its production started in September 2022, with a current daily gas production of over 43,000 cubic meters (Liu et al., 2024).

In September 2022, the first deep coalbed methane horizontal well in the LinXing area on the eastern edge of the Erdos Basin, named "Deep Coal 1," successfully started production with a test production rate of 60,000 cubic meters per day (Liu et al., 2024). Deep coalbeds expanded the development of unconventional energy. Compared to shallow coalbeds, deep coalbeds is more challenging, due to high temperature and high stress gradients, high degree of metamorphism, and high free gas content, which pose higher requirements on the development of deep coalbeds.

Horizontal drilling and hydraulic fracturing are used to effectively extracted the methane from CBM. Considering of deep burial, high structural stress, and poor physical properties of deep coal seams in China, extreme volume stimulation methodology was developed and implemented, with variable viscosity fracturing fluid system, slickwater for proppant suspension.

In this study a new HVFR was implemented in China deep coalbed methane in the LinXing and ShenFu area. Such HVFR can be customized to various water quality having a wide range of total dissolved solids values. This provides a flexibility to design fracturing jobs in unconventional reservoirs based on field conditions and the possibility to use the flowback instead of fresh water.

Fracturing concept and method

Deep CBM is different from shallow-middle coal seams, the fracturing concepts and targets are closer to those of shale fracturing. Conventional hydraulic fracturing—reliant on primary fracture propagation and gradual pressure depletion—demonstrates limited efficacy in deep coal seams (>2,000 m burial) due to elevated geo-stresses (> 20 MPa), impaired permeability (<0.36 mD), and pronounced plastic deformation. To overcome these constraints, an integrated extreme-volume stimulation methodology was developed, anchored on two core principles:

1. **Maximized Reservoir Contact (MRC):** Achieve comprehensive coal matrix disintegration to establish interconnected fracture networks, minimizing matrix-to-fracture flow resistance.
2. **Sustained Conductivity Preservation:** Ensure long-term fracture permeability through optimized proppant placement to maximize drainage volume and recovery efficiency.

The integrated extreme-volume fracturing methodology comprises six synergistic technical categories, each addressing specific subsurface challenges in deep CBM reservoirs (Yang et al., 2025):

Pretreatment. Pad-acid treatment (15% HCl): Dissolves acid-soluble minerals (calcite/dolomite/hematite) to reduce fracture initiation pressure and enhance near-wellbore conductivity.

Fluid System. Low-damage variable-viscosity slickwater (1–40 mPa·s): Balances proppant transport efficiency with fracture complexity generation while minimizing formation damage. Also requires good salt tolerance to meet the demand of recycle of flowback water.

Proppant Placement. Multi-size proppant (70/140–30/50 mesh) 3D placement*: Provides hierarchical support for primary/secondary fractures while mitigating embedment in plastic coals.

Perforation Strategy. Equal-aperture flow-limited perforation (16 holes/m): Ensures uniform cluster initiation and fracture geometry control; Cable-deployed soluble bridge plugs: Enables efficient stage isolation without post-frac intervention.

Stimulation Enhancement. Reduced-stage multi-cluster dense cutting: Maximizes fracture density per stage; Ball-sealer temporary diverter: Promotes complex fracture networks through forced fluid redistribution.

Stimulation Intensity. Ultra-high pumping rates (16–18 m³/min): Enhances fracture propagation and distal proppant placement; Extreme proppant volumes (≤ 28 m³/m): Maximizes stimulated reservoir volume (SRV) and long-term conductivity.

This integrated suite of technologies enables a paradigm shift from localized permeability enhancement to comprehensive volumetric reservoir transformation.

Laboratory Test Result

The HVFR system used in this work was tested extensively in laboratory prior to field implementation. A summary of the pertinent laboratory test results are described below:

Hydration Rate of HVFR system

Hydration rate is a critical parameter to consider, many of the guar gel and traditional friction reducer in the market are solid materials, which take longer time to hydrate, causing limitations in real-time wellsite utilization and often requires the use of specialized equipment to hydrate the polymers. Furthermore, dry solid polymer generates dust onsite leading to a negative impact on the environment and personnel health (Ghith et al., 2023; Dahlgren et al., 2018; Johnson et al., 2018; Van Domelen et al., 2017).

The hydration rate of the new HVFR system was found to be very fast, with complete hydration taking about 15 seconds as illustrated in Fig. 1. There is no requirement for a dedicated hydration process or equipment, thus saving equipment costs, time and well site space. No residual liquid or solid are produced after operation and no additional treatment is required. Fig. 1 further show that they hydration time is comparable using fresh water (TTW) or synthetic produced water (20,000 ppm and 60,000 ppm TDS water).

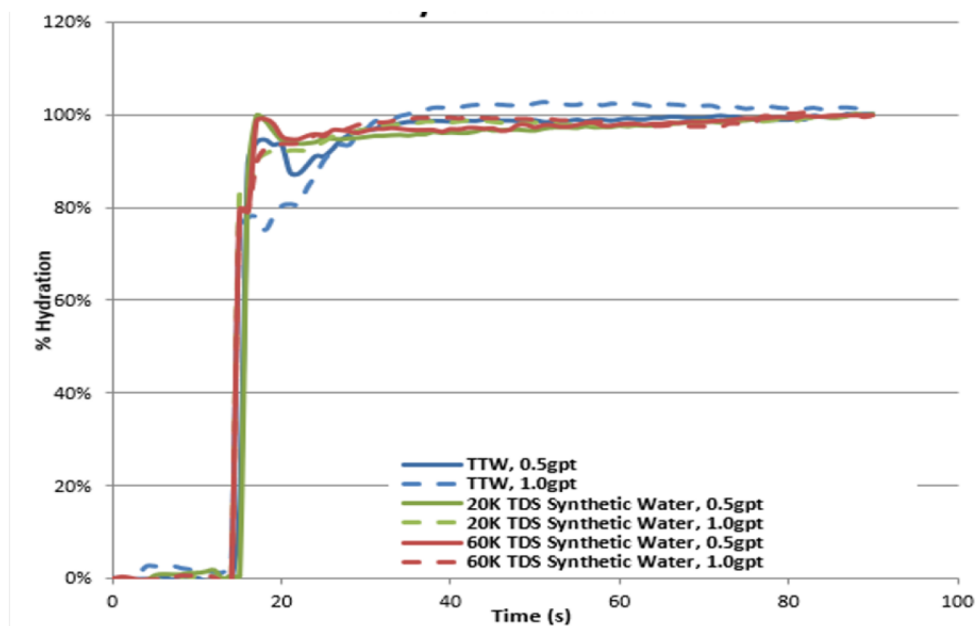


Figure 1—Hydration Rate of HVFR System in Various TDS Waters

Rheology Testing

Rheology testing at 230°F (110°C) was performed at various loadings in fresh water (200 ppm TDS), to ensure that the fluid temperature and shear resistance meet the normal operation requirements. Fig. 2, show the viscosity results. This system shows an excellent rheology profile, at concentration of 5 gpt the viscosity is almost 62 cP while it reached almost 220 cP at 8 gpt. It was found that viscosity values varied from 5–220 cP at 100 rpm shear rate for HVFR concentration from 1 to 8 gpt. High viscosity can be built up with low HVFR concentration.

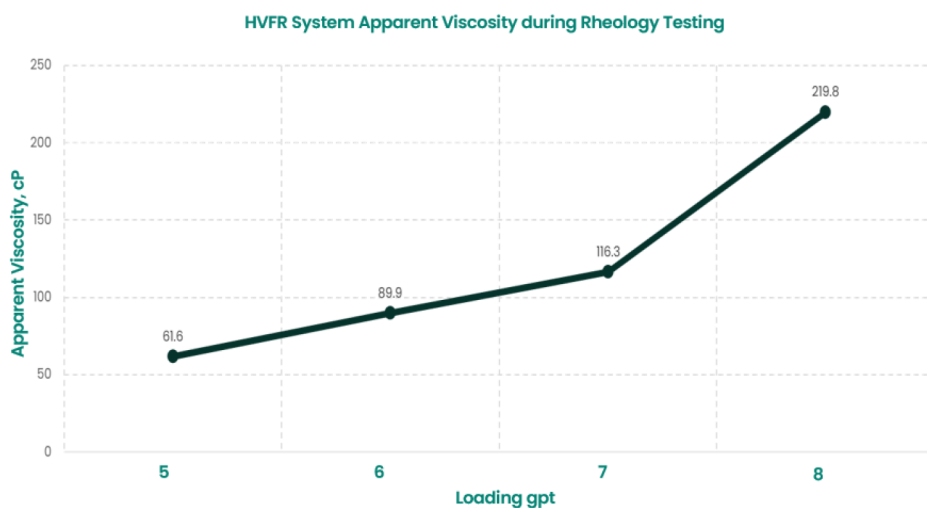


Figure 2—HVFR System Apparent Viscosity Results at 230°F as Function of Loading

Considering the field condition /water resources/flowback water utilization in Fig. 3 (typical contaminated, high turbidity flowback water with high content of calcium and magnesium), the given water quality could range from TDS 30k to TDS 170k as the statistics collected in Fig. 4 from LinXing and ShenFu Block in Ordos Basin, the HVFR system was mixed with high TDS water in the next step to generate viscosity profile.



Figure 3—Typical Flowback Water used in Fracturing Operations

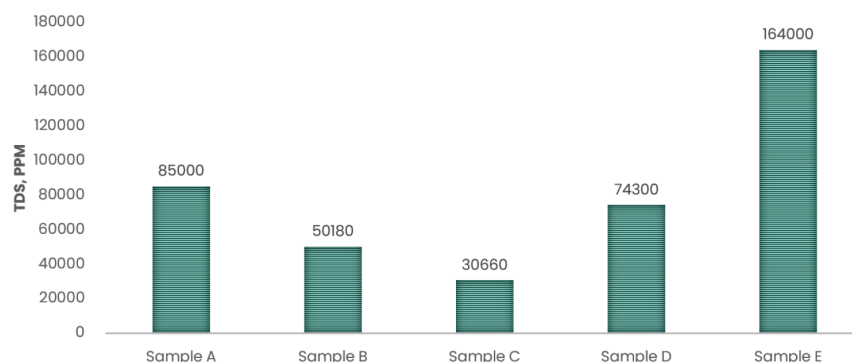


Figure 4—TDS Values of Flowback Water Samples Collected in Ordos Basin

Viscosity profile with TDS 253,300 ppm flowback water was recorded at room temperature and illustrated in Fig. 5. At such high TDS water, the viscosities are still high and vary between 5 to 60 cP for loadings from 1 to 10 gpt. The polymer can hydrate and build needed viscosity in produced water and the required viscosity can be achieved. This provides with flexibility for fracture job designs based on field condition.

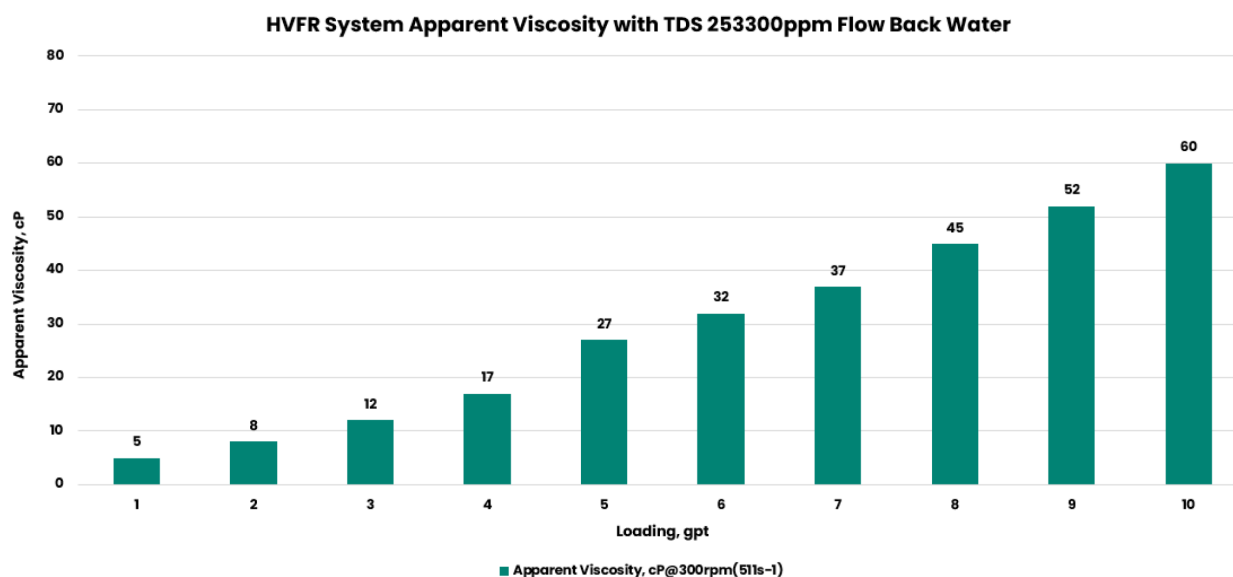


Figure 5—Viscosity Profile of Different HVFR Loading using 253,300 ppm TDS Flowback Water

Friction Reduction Ratio for Varying TDS Values

Friction reduction ratio of the HVFR system was measured for both fresh water and high TDS waters (20,000 and 60,000 ppm) to simulate various field conditions. It was found that friction reducing ratio was > 70 % for both fresh water and high TDS water with low HVFR dosage range from 0.5 to 1 gpt as shown in Fig. 6.

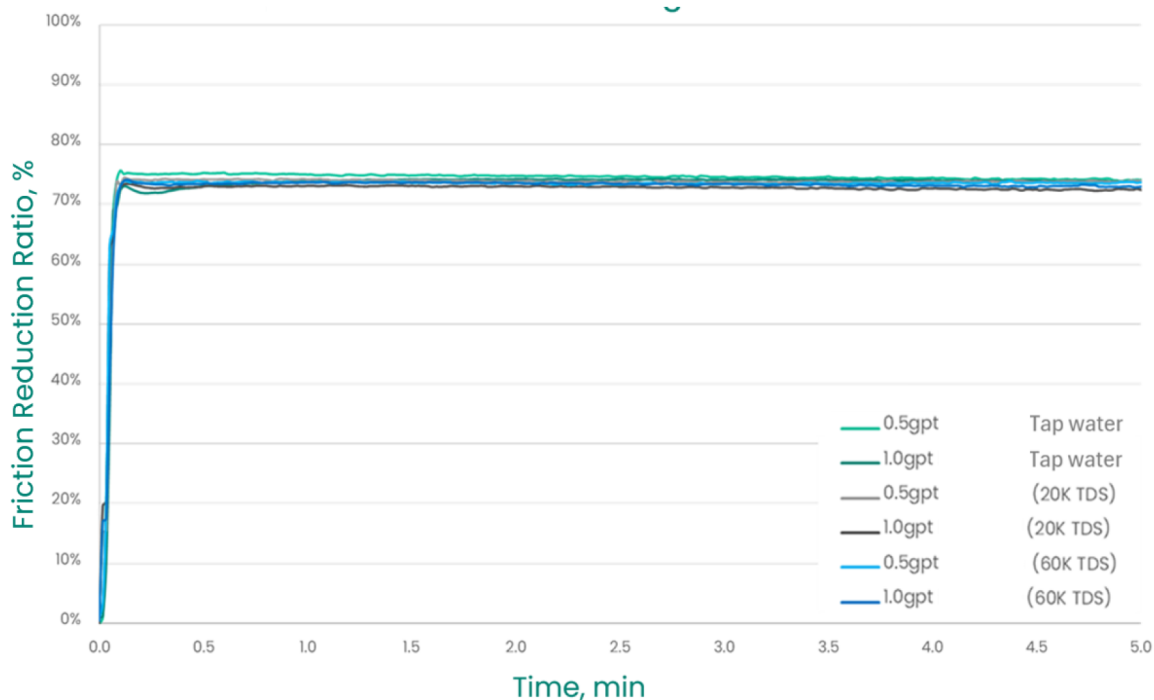


Figure 6—Friction Reducing Ratio @ Different TDS Values with 0.5 & 1 gpt HVFR Concentration

Friction reduction ratio was also measured using this new HVFR system during a Shale gas job (well XX1H) using produced water with TDS > 200k ppm conditions, as shown in Fig. 7 the friction reduction is higher than 75% (range from 75% to 85%) in each stage.

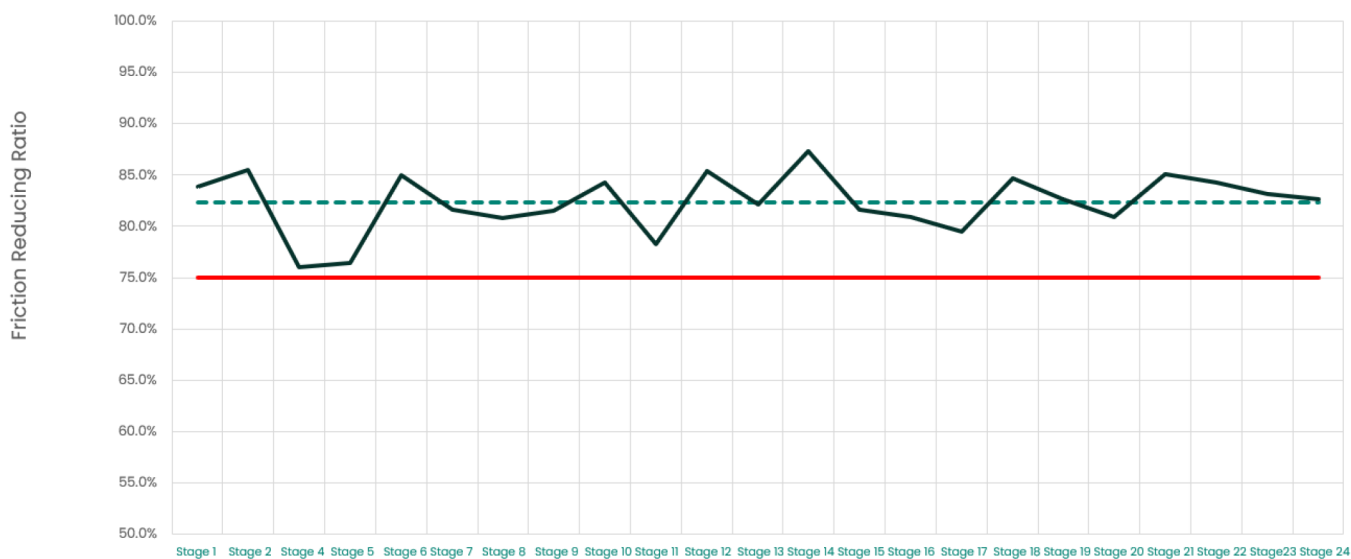


Figure 7—Friction Reducing Ratio for Well XX1H

Field Implementation

Considering the deep coal bed methane fracturing method and concept, HVFR system were utilized as trial assessment in LinXing and ShenFu Block, Ordos Basin, China. Many challenges were faced during this implementation, including:

Water scarcity. Large-scale matrix fracturing consumes a large amount of fresh water and brings environmental issues due to the large amount of flowback water generated, which contains chemicals and toxic components that could potentially contaminate underground water resources (). Operators face a dramatical increase of water demand, although it is significantly smaller than that irrigation and municipal use (D. D. Schmidt et al.,2025), alternatives can alleviate future increase water usage in unconventional reservoir hydraulic fracturing. The transportation and treatment cost of flowback water is high. Using flowback water can alleviate the need of fresh water and decrease cost. Even though the reuse of flowback water is not a new concept, the high salinity reservoir conditions were very challenging.

Long lateral horizontal well with higher proppant carrying requirement. Most deep coalbed methane wells in China are drilled horizontally through the expected pay zone. The main purpose of fracturing is to help successfully place a certain amount of proppant per length while making the best use of the horizontal section, extending fractures as far into the formation as possible with complex geometry, ultimately reach MRC (Maximized Reservoir Contact), and, in return having easier and faster gas recovery. Traditionally, proppant placement in long laterals has been achieved using guar gum or other gel based fracturing fluids, but the residues left after fracturing fluid is broken could be a problem causing formation damage and lead to reduction in regained conductivity. Operators focused on minimizing conductivity loss and formation damage now prefer to use slickwater systems. As proppant concentration and laterals length increase, the capability of traditional slickwater to transport proppant becomes limited, due to the polymer's inability to properly hydrate and remain its stability in high salinity water. Proppant carrying capacity of traditional slickwater is limited and greatly affected by water quality.

The operator's high-level objectives include reducing surface treating pressure, as well as carrying and embedding larger mesh proppants at high concentrations in varying conditions of flowback water and produced water.

Results in Field Application

HVFR system can build significant viscosity and elasticity which gives the ability to not only carry high concentration proppant but also larger mesh sizes proppant deep into the formation in poor water quality.

HVFR system was able to effectively place larger mesh size proppants at higher concentrations using high TDS flowback water, with controllable surface treating pressure and strong fluid stability, where a typical operation pressure profile as shown in Fig. 8. More than 60 stages treatment help operator save 60,000 m³ fresh water. The HVFR system exceeded financial savings, and timing demands by pumping using less fresh water and place larger and higher concentration proppant (3.5 ppa) to achieve fracture conductivity. The HVFR successfully carried high concentration proppant (from 70/140 to 40/70 mesh, 30/50 mesh), leading to higher conductivity fractures and better production performance.

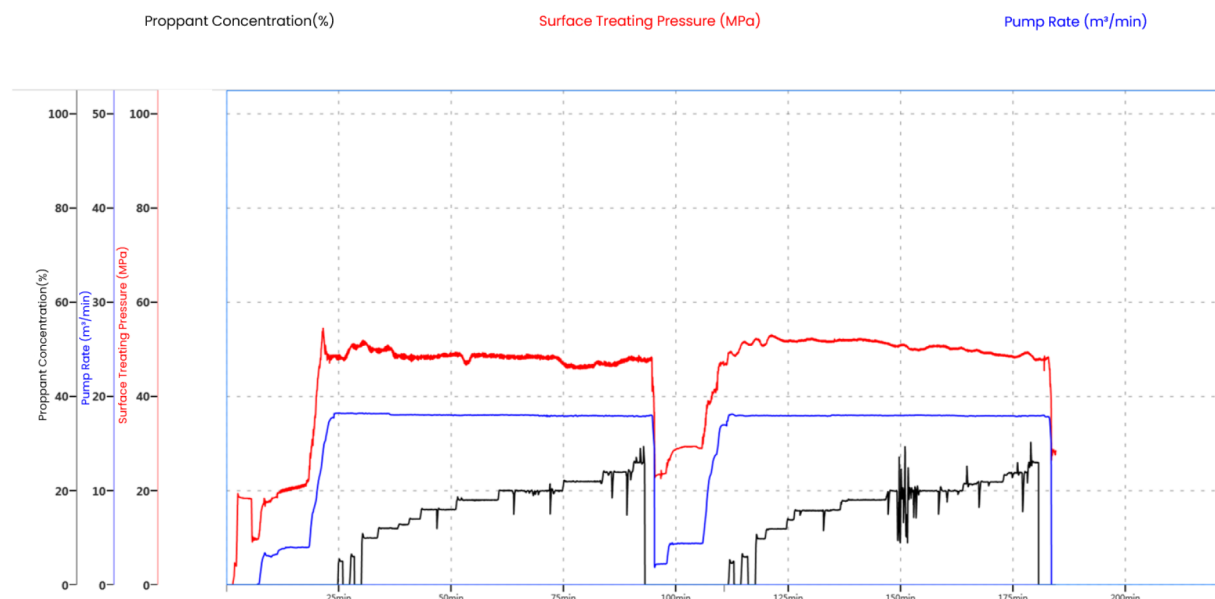


Figure 8—Typical Operation Pressure Profile using HVFR System with High TDS Flowback Water

Result Comparison

The HVFR system was used in several horizontal wells where more than 60 stages were pumped with flowback water in LinXing and ShenFu block located in Ordos Basin. Compared with traditional slickwater, HVFR system successfully solve the problem of utilization of produced water and flowback water with high salinity (TDS), alleviated the pressure of water scarcity and save the cost of flowback water treatment and transportation. Compare with guar or polymer gel (other high viscosity fracturing fluids), HVFR system have the ability to achieve desired fracture complexity with low formation damage, with superior friction reduction, contribute to fracture conductivity and proppant placement. It was found to perform exceedingly well for high TDS water as compared to other fracturing fluids especially with high ability to place high sand concentrations. During the treatment it was found that the pressure fluctuation is relatively small, with little difference compared to HVFR mixed with fresh water. Two scenarios were listed below to indicate the exceeding performance HVFR system than other fracturing fluids and strong stability during mixed with high salinity water.

Application 1: HVFR System Compared with Other Fracturing Fluids

In this application, HVFR system mixed with flowback water was investigated and compared with other fracturing fluids from other suppliers mixed with same flowback water. Taking Platform B in LinXing Block as example, 6 offset horizontal gas wells were fractured with different fracturing fluids, within the same target formation, No.9 coal seam (Benxi Formation). The six wells have similar well profiles, total depth, TVD and completion design, the only difference is the fracturing fluid. Three fracturing fluids (HVFR, Frac Fluid A and Frac Fluid B) were used, each tested in two wells.

To determine the success of the technology during the pumping of the treatment, various parameters were recorded and analyzed. Pressure fluctuation analysis was conducted by calculating the difference between each minute recorded pressure data and calculated average pressure data with the same pump rate. The pressure difference curve shows the pressure fluctuation within the treatment, which also indicates the fluid stability. The closer the pressure fluctuation ($(\text{Pressure} - \text{Average Pressure}) / \text{Average Pressure}, \%$) to 0, the better the fluid performance and treatment execution.

Fig. 9 and 10 shows the pressure fluctuation analysis of Frac Fluid A and Frac Fluid B fracturing fluid. Fig. 11 shows the HVFR system pressure fluctuation analysis. The overall pressure fluctuation of HVFR

system treatment is relatively small compared to the other two fracturing fluids, indicating that HVFR system performance is relatively stable.

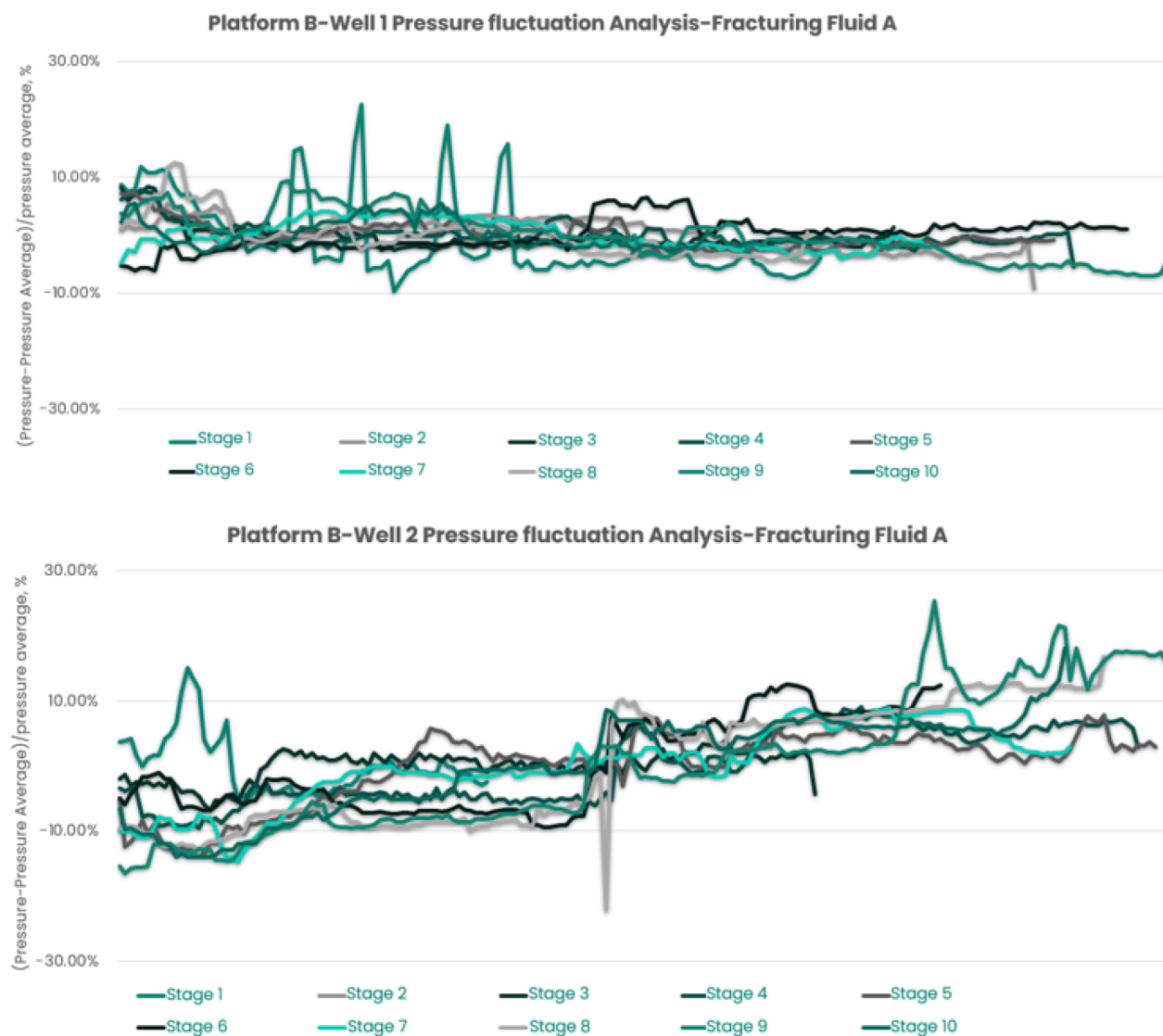


Figure 9—Pressure Fluctuation Analysis for Wells using Fracturing Fluid A

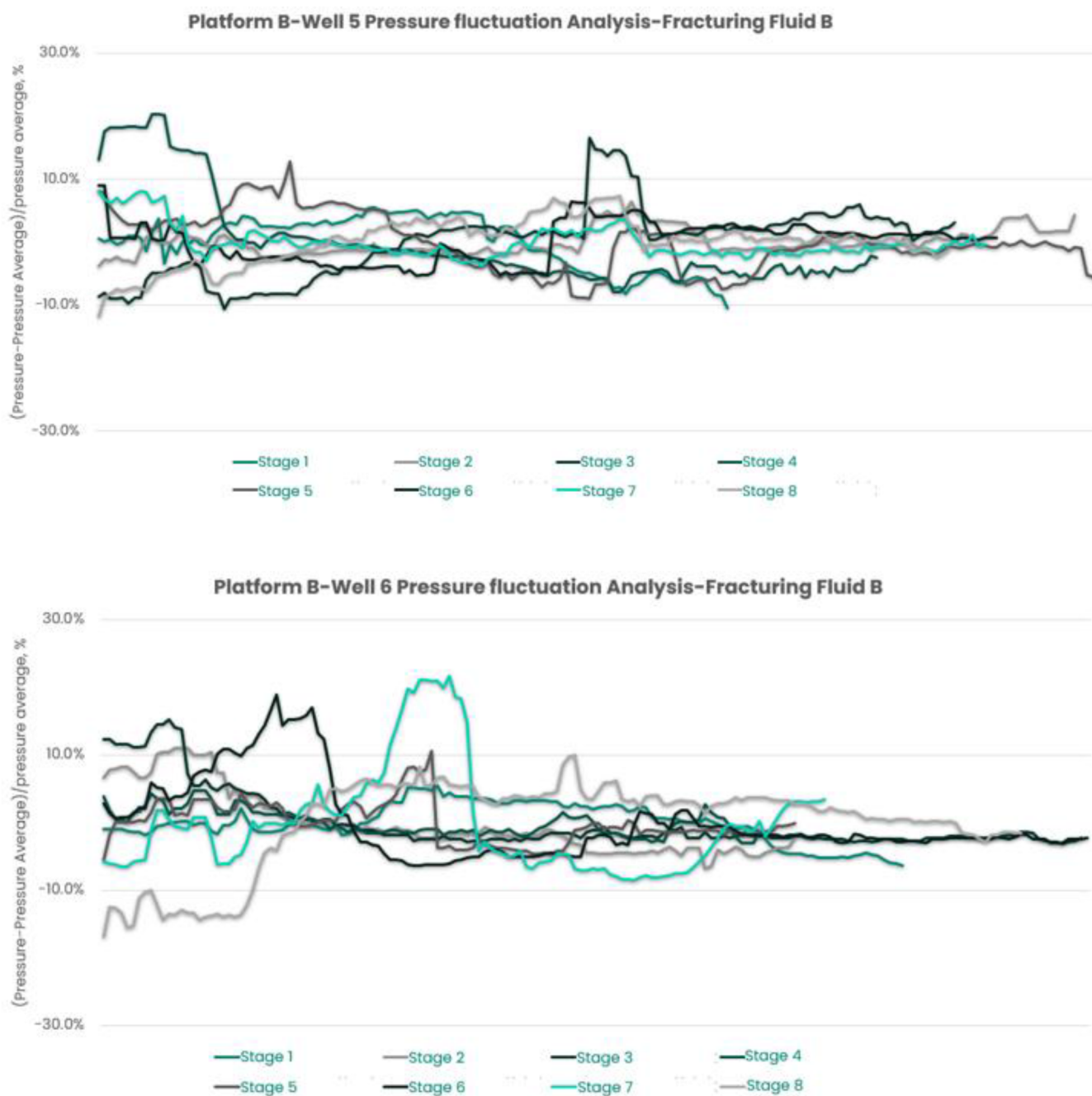


Figure 10—Pressure Fluctuation Analysis for Wells using Fracturing Fluid B

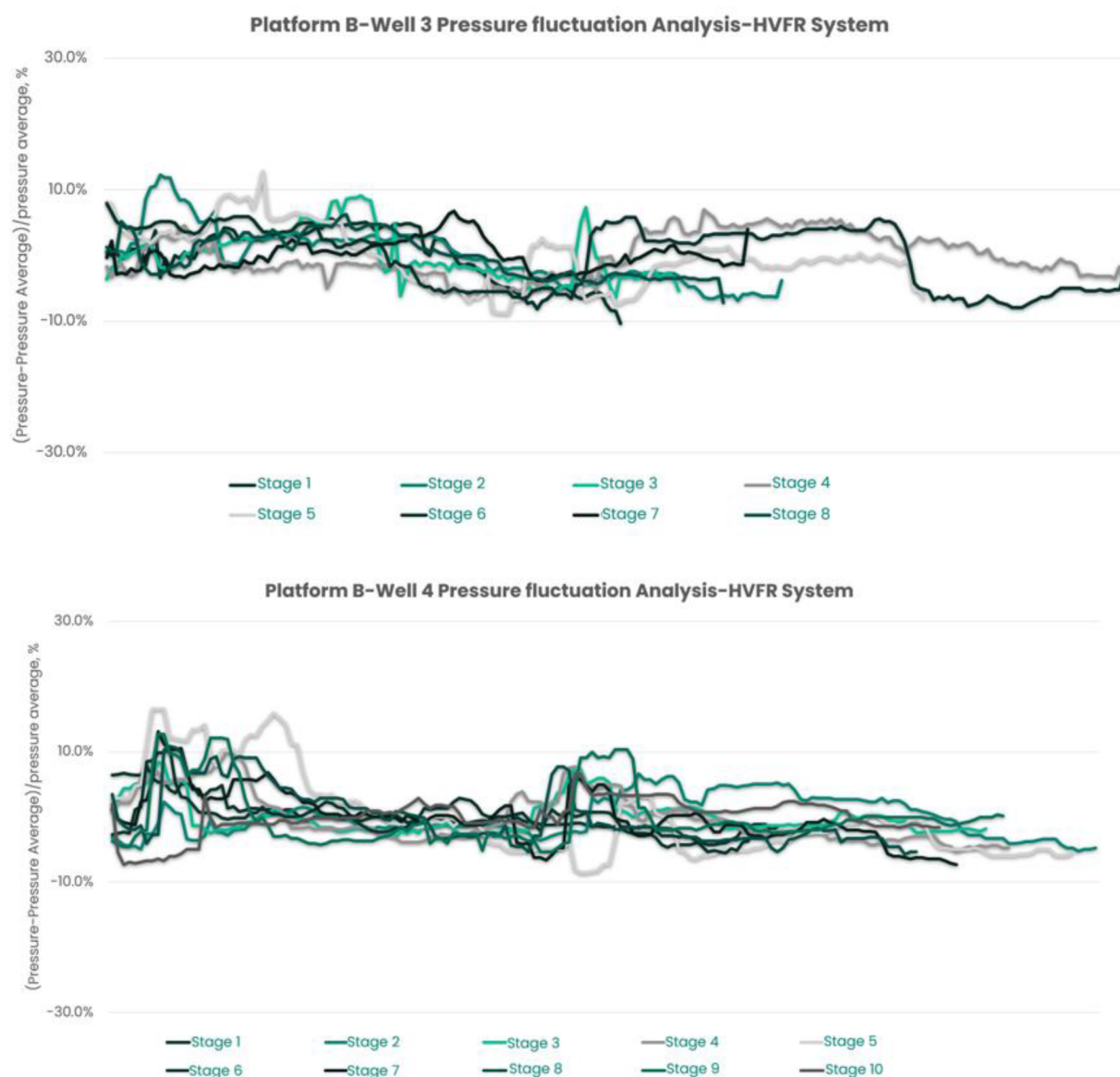


Figure 11—Pressure Fluctuation Analysis for Wells using HVFR System

It is worth mentioning that the high concentration proppant transport ability of HVFR system, the pressure curve shows a downward trend, indicating a smooth proppant placement during high concentration, contributes to fracture expansion space and fracture network. In contrast, supplier A's pressure curve shows an upward trend which may reflect the poor performance of fluid in early stage make it difficult for later proppant carrying, which is unfavorable for the overall fracture geometry and complexity.

Based on this primary pressure fluctuation analysis, more treatment data were collected from LinXing and ShenFu Block. Standard deviation for pressure data was calculated to furthermore indicate fluid stability. A low standard deviation indicates that the values are clustered closely around the average, while a high standard deviation means the values are more dispersed. In this case, smaller standard deviation reflects smaller pressure fluctuation. As shown in Table 1, HVFR system has the smallest standard deviation compared to the other two fracturing fluids, further confirmed the feasibility of using HVFR system with flowback water for large volume matrix fracturing.

Table 1—Standard Deviation of Pressure Fluctuation Analysis

Fracturing Fluid	Well Name	Pressure Standard Deviation	Average Standard Deviation
Fracturing Fluid A	Platform B-Well 1	1.65	2.2
	Platform B-Well 2	3.54	
	Platform A-Well 1	2.06	
	Platform A-Well 2	1.46	
HVFR System	Platform B-Well 3	2.12	1.9
	Platform B-Well 4	2	
	Platform A-Well 3	1.31	
	Platform A-Well 4	2.33	
Fracturing Fluid B	Platform B-Well 5	2.26	2.44
	Platform B-Well 6	2.52	
	Platform A-Well 5	2.47	
	Platform A-Well 6	2.51	

Application 2: Flowback Water vs. Fresh Water

In this application, HVFR treatment fluid that based on flow back water was investigated and compared with HVFR fluid based on fresh water. A horizontal well operated in LinXing block was hydraulically fractured in 10 stages. Flowback water was used in stage 4 while fresh water was used in the other stages. Both achieved 100% proppant carrying with good fluid stability. The difference is the HVFR concentration to achieve the needed viscosity in the various stages of the treatment. Although fresh water contributes to lower chemical concentration, flowback water mixed with higher concentration of HVFR meets the requirement.

As show in Fig. 12, pressure profile of stage 4 (HVFR mixed with flowback water) compared to other stages (HVFR mixed with fresh water), both went smoothly and have a downward trend indicating good fracture extension and proppant embedment.

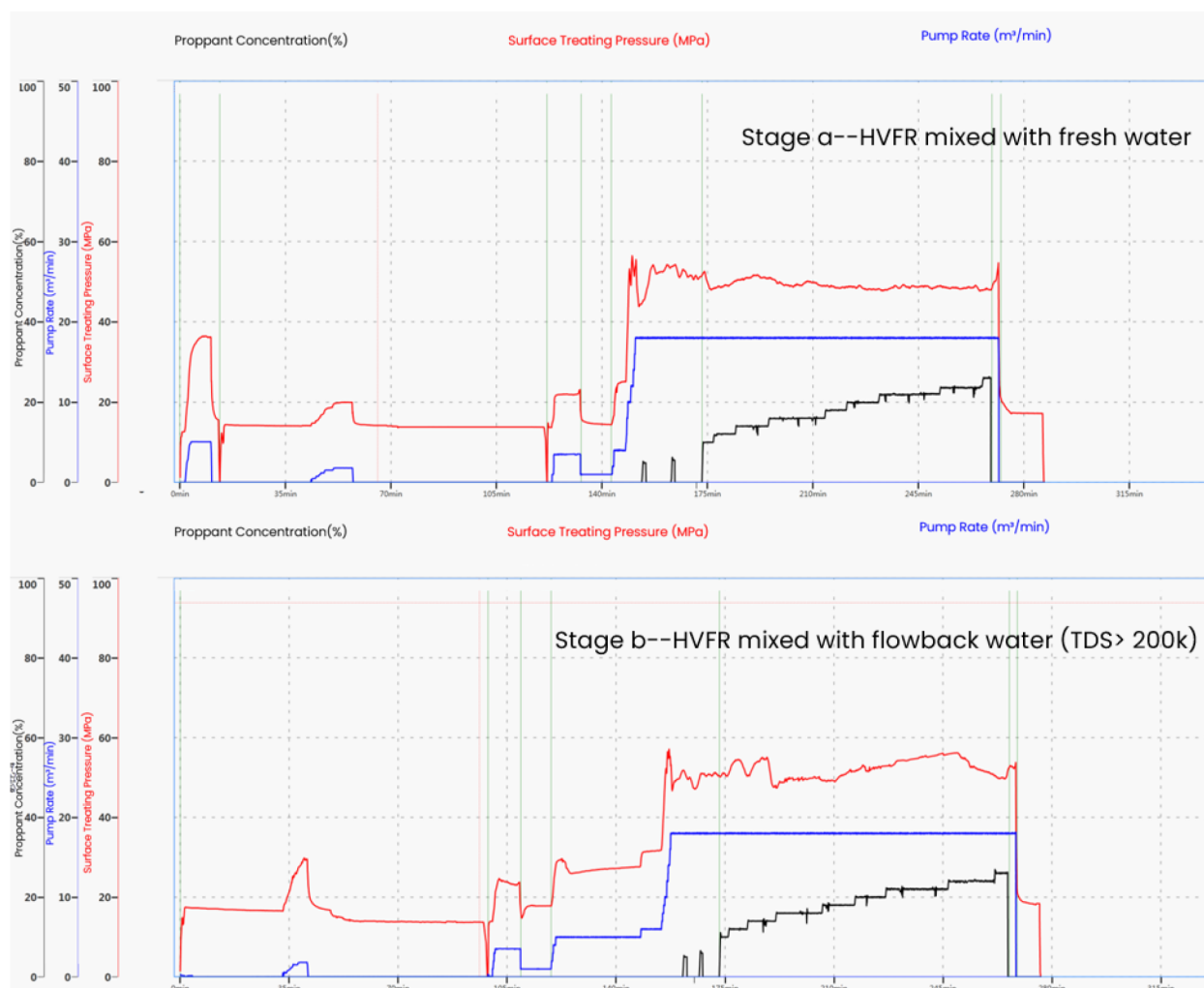


Figure 12—Operation Pressure Curves Comparison of HVFR mixed with Fresh Water and Flowback Water

Using flowback water allowed very successful operation. The HVFR built enough viscosity to provide with the desired geometry and place efficiently the proppant. This greatly alleviates the current situation of water scarcity in LinXing and ShenFu block in China.

Conclusion

This paper introduced a new HVFR system that is applicable to both fresh and high TDS produced water. The field application shows that the wells used the HVFR system show excellent performance with 100% proppant transportation, smooth pressure curves and strong fluid stability during treatment. These fluids are very well suited for CBM, shallow and deep, applications and do not leave any residue behind, decreasing the risk of formation damage due to not fully broken fluid.

Superior Fluid Performance. The HVFR achieves sufficient friction reduction and rapid hydration across a wide salinity range, enabling efficient proppant transport even in high salinity flowback water. It builds tunable viscosity with low concentrations, maintaining stability under high temperature/shear conditions.

Field Success in Challenging Conditions. Deployed in >60 stages across Ordos Basin horizontal wells, the system enabled 100% proppant placement using high TDS flowback water, saving freshwater usage and reducing treatment costs. It outperformed traditional slickwater and polymer gels by delivering higher proppant concentrations, complex fracture geometry, and minimal formation damage.

Operational and Environmental Advantages. Eliminated freshwater dependency while handling high turbidity, saline flowback water. Showed exceptional fluid stability, evidenced by low pressure fluctuations and smooth fracture extension.

Technical Validation. Pressure fluctuation analysis confirmed superior stability over competing fluids, ensuring consistent fracture propagation and proppant embedding. The chemistry enables cost-effective use at higher loadings, making it viable for large-scale applications.

Broader Impact. This HVFR system establishes a sustainable pathway for fracturing unconventional reservoirs, reducing environmental footprint while enhancing productivity in water-scarce regions. Its adaptability to diverse water qualities positions it as a transformative solution for global deep CBM and shale developments.

Nomenclature

HVFR	High viscosity friction reducer
TDS	Total dissolved gas
gpt	gallon per thousand gallons
MRC	Maximized reservoir contact

References

- H. Liu., C. Zou., S. Yin. et al 2024. Formation, distribution, sweet spot evaluation and prospect of coal-measure gas in China[J]. *Natural Gas Industry*, 2024, **44**(x): to be published.L. Zhou, X. Li, Y. Liu, H. Ding. et al., 2025. Revolutionary breakthroughs and key theories and technologies in deep coalbed methane development[J]. *Natural Gas Industry*, 2025, 45(05): 17-30. DOI: [10.3787/j.issn.1000-0976.2025.05.002](https://doi.org/10.3787/j.issn.1000-0976.2025.05.002).
- F. Yang, B. Li, and Z. Hao. Deep Coalbed Methane Reservoir Simulation Technology In Shenfu Block. Paper presented at the International Petroleum Technology Conference, Kuala Lumpur, Malaysia, 18 – 20 February 2025. DOI [10.2523/IPTC-24742-MS](https://doi.org/10.2523/IPTC-24742-MS).
- Ghith, B., Shari, Dunn N. et al 2023. Impact of HVFR Loading Concentrations on its Viscoelastic Properties and Proppant Transport Performance. Paper presented at the SPE Eastern Regional Meeting, Wheeling, West Virginia, USA, October 2023. DOI [10.2118/SPE-215915-MS](https://doi.org/10.2118/SPE-215915-MS)
- Dahlgren, K., Green, B., Williams, B. et al 2018. Case Studies of High Viscosity Friction Reducers HVFR in the STACK Play. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, 23-25 January. SPE-189893-MS. <https://doi.org/10.2118/189893-MS>
- Johnson, M., Winkler, A., Aften, C. et al 2018. Successful Implementation of High Viscosity Friction Reducer in Marcellus Shale Stimulation. Paper presented at the SPE/AAPG Eastern Regional Meeting, Pittsburgh, Pennsylvania, USA, 7-11 October. SPE-191774-18ERM-MS. <https://doi.org/10.2118/191774-18ERM-MS>.
- Van Domelen, M., Cutrer, W., Collins, S. et al 2017. Applications of Viscosity-Building Friction Reducers as Fracturing Fluids. Paper presented at the SPE Oklahoma City Oil and Gas Symposium, Oklahoma City, Oklahoma, USA, 27-31 March. SPE-185084-MS. <https://doi.org/10.2118/185084-MS>
- D.D. Schmidt; G. Harms; A. Thiel; J. Neubeker; T. Hopfauf. 2025. Recent Developments and Utilization of Produced Water in Bakken Well Stimulations. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, February 2025. SPE-223512-MS/ <https://doi.org/10.2118/223512-MS>